Quantification: 872_wt_hp_z04
==============================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	13
interior lacuna count	10
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 10)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	1940.9	1853.5	1010.93	px^2
major_axis_length_px	lacuna shape	75.8307	76.683	24.3617	px
minor_axis_length_px	lacuna shape	33.9653	31.7232	9.4154	px
aspect_ratio	lacuna shape	2.2324	2.3906	0.5464	unitless
eccentricity	lacuna shape	0.8623	0.9076	0.1015	unitless
solidity	lacuna shape	0.7719	0.7755	0.064	unitless
roots_count	headline	15.1	15.5	4.8408	count
ring_length_r30_px	headline	680.1	762	194.984	px
ring_length_r60_px	reported	1606.5	1752.5	395.107	px
owned_length_px	ownership-dependent	3376.99	3292.33	1279.95	px
edge_count	ownership-dependent	219	210.5	88.7994	count
mean_edge_length_px	ownership-dependent	15.6534	15.8783	1.6297	px
perimeter_px	normalised	254.63	264.764	86.6163	px
ring_area_r30_px2	normalised	9022	9101.5	2028.2	px^2
ring_area_r60_px2	normalised	22120.7	22141.5	4248.88	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0752663	0.0790986	0.0121753	px^-1
ring_density_r60	normalised	0.0731669	0.0744297	0.0139928	px^-1
roots_per_100px_perimeter	normalised	6.1064	6.1795	1.2747	per 100 px
ring_attached_length_r30_px	network v2	648.7	712.5	202.532	px
ring_attached_length_r60_px	network v2	1492.2	1697.5	460.387	px
ring_length_w_r30_px	network v2	778.366	874.047	224.674	px
ring_length_w_r60_px	network v2	1830.06	2000.51	464.498	px
sholl_crossings_r10	network v2	14.5	15	4.6007	count
sholl_crossings_r20	network v2	16.1	16.5	4.6774	count
sholl_crossings_r30	network v2	20.9	21	6.6408	count
loop_count	loops, report only	35.3	33.5	22.9495	count
loop_edge_share	loops, report only	0.5552	0.6115	0.1394	share
ring_width_mean_r30_px	width	7.2107	7.2026	0.4206	px

Per field
----

measure	value	units
analysed_area_px2	1.02096e+06	px^2
total_skeleton_length_px	39230	px
canalicular_length_density_per_px	0.0384244	px^-1
junction_count	1415	count
junction_density_per_px2	0.00138594	px^-2
lacuna_count	13	count

lacunae_per_px2	1.273e-05	px^-2
skeleton_component_count	137	count
mean_component_length_px	286.35	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0435228	px^-1
field_density_without_flagged_per_px	0.0429989	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	53	px
unattached_component_count	4	count
canalicular_length_density_attached_per_px	0.0384244	px^-1
field_length_density_with_unattached_per_px	0.0384763	px^-1
loop_count	487	count
loop_edge_share	0.6139	share
edge_count	2748	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7453	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	37238	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3025	100.943	44.409	2.273	0.898	0.65	-1.4641	211.119	126.158	False
2	2144	87.4647	34.0354	2.5698	0.9212	0.7528	-1.2557	226.871	777.66	False
3	2841	103.213	37.4578	2.7554	0.9318	0.8351	-1.3658	276.205	420.61	False
4	497	30.0795	23.8394	1.2618	0.6098	0.7827	-0.6524	264.441	629.968	False
5	3125	68.6713	60.8024	1.1294	0.4648	0.909	0.622	332.681	995.111	True
6	3954	114.494	48.2385	2.3735	0.9069	0.7517	-1.4049	393.248	55.5005	True
7	3573	94.8523	52.0807	1.8213	0.8358	0.8284	-1.4851	395.903	280.417	False
8	1409	76.4593	28.6406	2.6696	0.9272	0.7683	-1.5183	392.216	831.581	False
9	2226	76.9066	39.7827	1.9332	0.8558	0.7987	-1.558	506.567	185.868	False
10	1503	75.7312	26.0285	2.9095	0.9391	0.8628	-1.3051	568.045	702.943	False
11	1123	61.8357	30.8107	2.007	0.867	0.6785	-1.3397	588.985	20.1576	True
12	628	38.8866	23.9678	1.6225	0.7875	0.7235	-1.4785	619.033	332.451	False
13	1563	73.771	29.411	2.5083	0.9171	0.717	-1.5252	651.639	225.151	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	23	955	1991	3217.05	203	15.8475	7.1991
2	19	770	1744	2739.41	167	16.4036	7.1819
3	14	754	1880	3468.16	218	15.909	7.206
4	9	449	1227	3818.33	237	16.1111	7.8128
5	13	358	852	1304.12	86	15.1641	7.5825
6	17	607	1362	1370.58	94	14.5807	7.3359

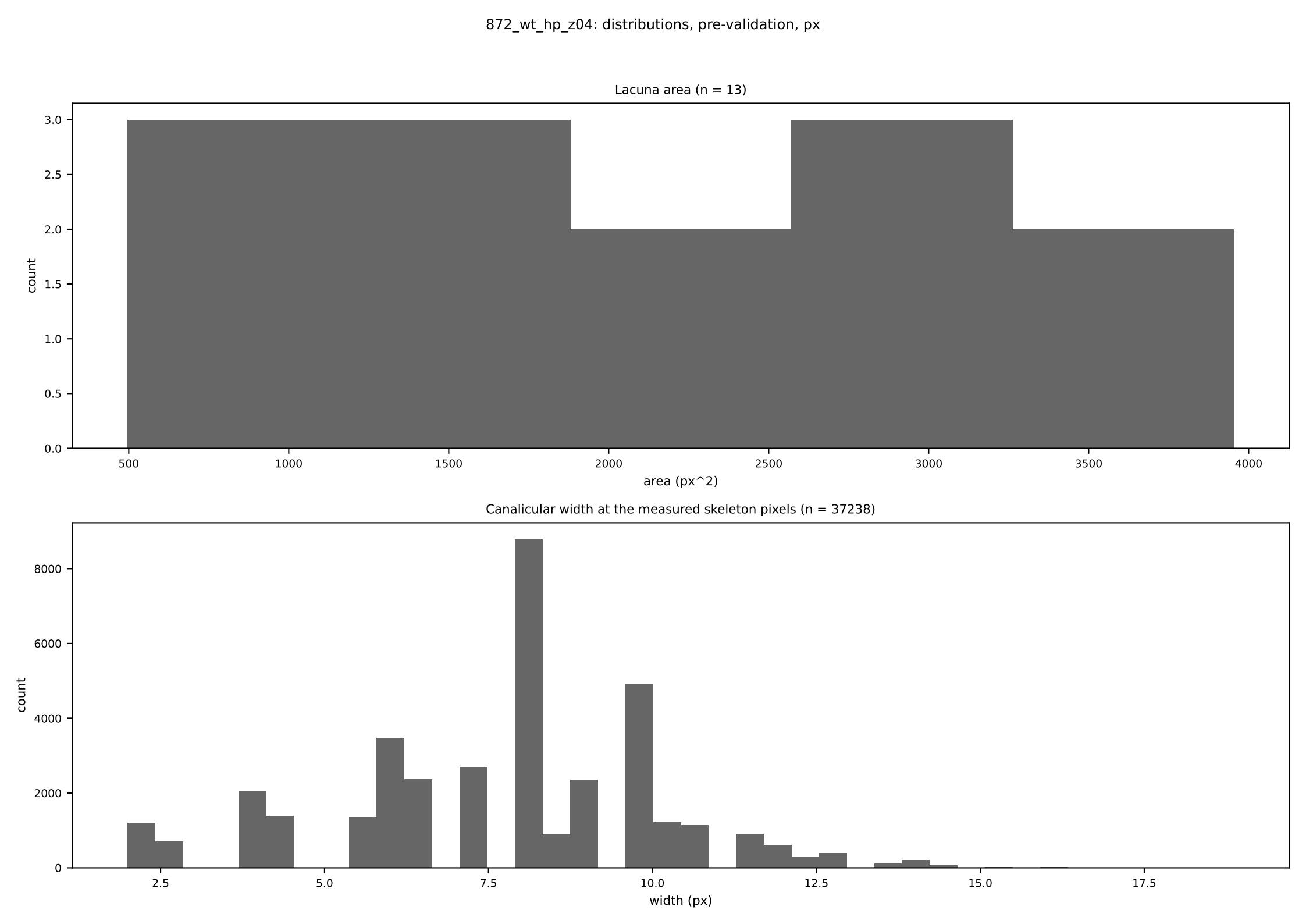
7	17	874	2140	4649.21	306	15.1935	7.846
8	12	409	1137	2198.56	126	17.4489	6.4577
9	20	783	1870	2108.23	171	12.3288	7.3483
10	9	506	1063	2047.17	113	18.1165	6.8757
11	12	482	1240	2316.1	155	14.9426	6.8827
12	11	499	1252	6156.17	413	14.906	6.8692
13	17	802	1761	3367.6	236	14.2695	7.3103

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	425.345	11873	28650	1	1	0.0804346	0.0694939	5.4074
2	273.321	9846	24205	1	1	0.0782044	0.0720512	6.9515
3	288.978	10581	26042	1	1	0.0712598	0.0721911	4.8447
4	117.054	5613	16004	1	1	0.0799929	0.0766683	7.6888
5	236.551	6707	15888	0.711	0.6585	0.0533771	0.0536254	5.4956
6	432.996	10444	22425	0.8508	0.7745	0.0581195	0.0607358	3.9261
7	315.191	11229	25289	1	1	0.0778342	0.0846218	5.3936
8	256.208	8359	21924	1	1	0.0489293	0.051861	4.6837
9	273.764	9314	21493	1	1	0.084067	0.0870051	7.3055
10	204.445	8382	22359	1	1	0.0603675	0.0475424	4.4022
11	216.752	5820	13951	0.7153	0.655	0.0828179	0.0888825	5.5363
12	148.61	6134	15204	1	1	0.0813499	0.0823467	7.4019
13	243.386	8889	20037	1	1	0.0902239	0.0878874	6.9848

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	930	1748	1101.89	2270.4	22	18	26
2	693	1661	886.867	1995.75	17	16	26
3	732	1812	863.654	2153.24	18	19	22
4	448	1150	530.588	1400.36	9	13	11
5	308	675	399.635	969.708	7	12	12
6	564	1127	706.683	1562.71	14	14	14
7	854	2137	998.767	2464.55	19	25	31
8	342	940	458.605	1271.38	9	12	13
9	779	1853	884.441	2140.3	14	17	27
10	459	740	574.132	1183.22	11	8	15
11	475	1239	544.676	1418.12	12	12	14
12	466	1147	565.132	1416.15	10	14	20
13	784	1734	919.583	2005.28	16	19	18


872_wt_hp_z04: distributions, pre-validation, px
Lacuna area (n = 13)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 37238)
count
width (px)